



Req2UI

From Natural Language to Secure Software Artifacts Using AI

Arab Academy for Science, Technology & Maritime Transport (AASTMT) — College of Computing and Information Technology, Smart Village — 2025/2026

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What we'll cover

Eight sections, ~20 minutes.

01

Motivation & Problem

Why manual SRS fails.

02

Objectives & Related Work

What we set out to do.

03

Proposed Solution & Architecture

The big idea.

04

The 10-Stage Pipeline

Centerpiece.

05

Concurrency, Resumability & Security

How it stays fast & safe.

06

UML, UI Code & Traceability

Outputs.

07

Tech Stack & Live Demo

What's running.

08

Testing, Results & Future Work

Validation + roadmap.

Why Req2UI?

Manual SRS authoring fails the modern team.



01

SLOW

Requirements authored by hand take days to weeks per project. Iteration is painful and feedback loops are long, especially when stakeholders change their minds mid-cycle.



02

INCONSISTENT

Style, depth, and terminology vary across teams and across projects. Two engineers reading the same input will produce two very different SRS documents.



03

INSECURE

Security requirements are routinely missing or tacked on late. Teams treat them as an afterthought rather than a first-class deliverable of the spec phase.

The tooling gap

Today's tools each cover only one slice — none span the whole flow.



A

Requirements tools

Generate an SRS document and stop there. No downstream artifacts, no security generation, no traceability to tests or UI.



B

UI prototyping tools

Produce screens and components, but assume requirements are already settled upstream. No link back to a spec.



C

Test automation tools

Assume requirements exist. They generate tests from an external spec or from existing code — not from natural language.



No tool auto-generates security requirements. No tool keeps end-to-end traceability across artifacts.

Five objectives

What Req2UI set out to deliver.

01



Single input

Accept a plain-language description of a software system as the only input. No structured forms, no DSL.

02



Full SRS

Auto-generate a complete IEEE-830 SRS with functional, non-functional, and security requirements.

03



Test cases

Generate functional test cases aligned to IEEE 829 and security test cases driven by OWASP Top 10 (2021).

04



UML + UI

Produce UML diagrams (use-case, class, sequence) and runnable UI code (HTML + Tailwind) per screen.

05



Traceability + export

Maintain end-to-end traceability across artifacts and export the bundle as PDF, DOCX, LaTeX, and CSV.

What exists, what's missing

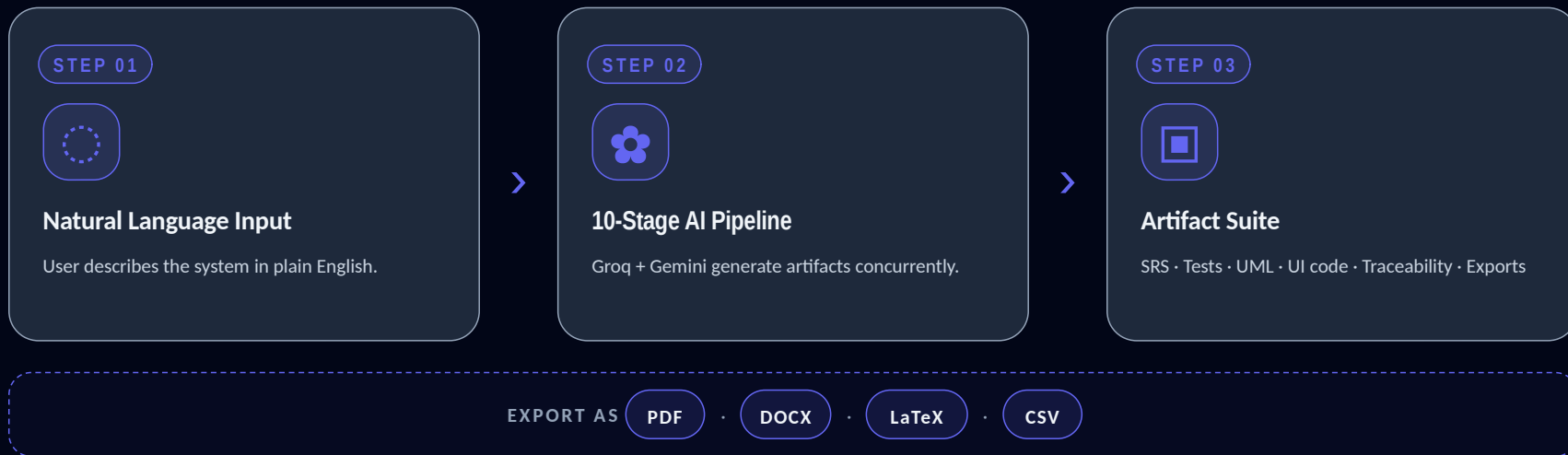
Today's categories each cover one slice. Req2UI spans all five.

CATEGORY	REQUIREMENTS	UI PROTOTYPE	TESTS	SECURITY	TRACEABILITY
Requirements generators	✓	✗	✗	✗	✗
UI prototypers	✗	✓	✗	✗	✗
Test automators	✗	✗	✓	✗	✗
Security scanners	✗	✗	PARTIAL	✓	✗
Req2UI OURS	✓	✓	✓	✓	✓

First integrated NL → SRS → UI → functional + security tests pipeline with auto security generation and full traceability.

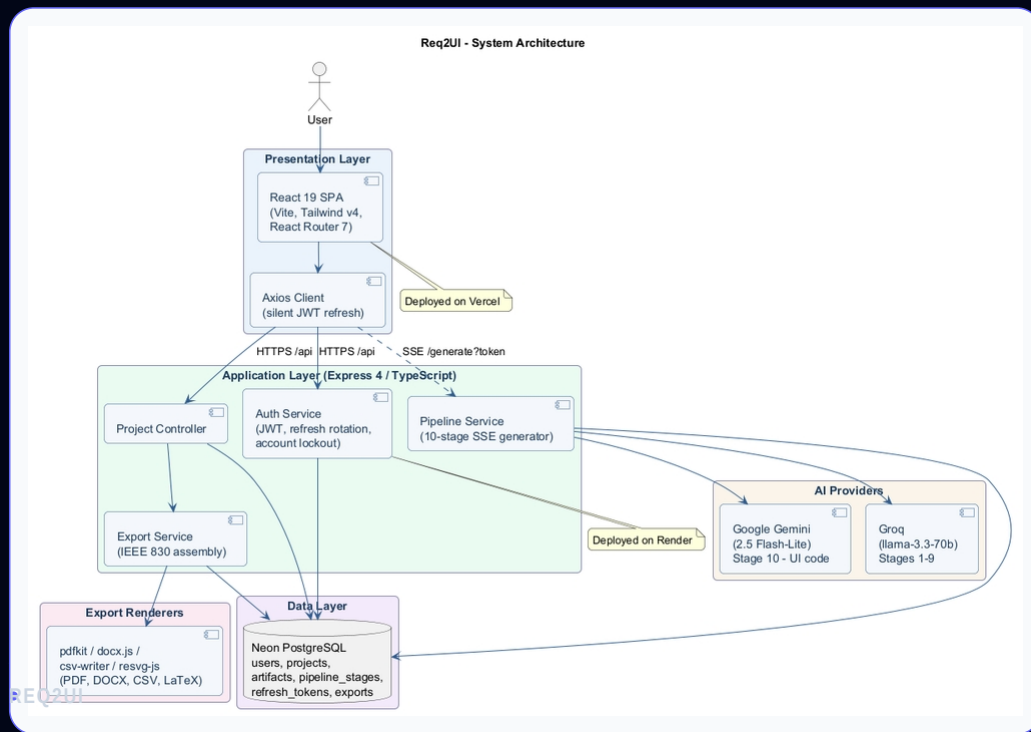
One input. A complete, security-aware artifact suite.

A single natural-language description drives a 10-stage AI pipeline that emits every artifact a spec phase needs.



Layered architecture

React SPA → Express API → Groq / Gemini → Neon Postgres.



PRESENTATION

React 19 SPA

TypeScript app with Tailwind v4, Vite, React Router 7. SSE listener streams live pipeline progress.



APPLICATION

Express 4 API

TypeScript service orchestrating the 10-stage pipeline. JWT auth, Zod validation, Helmet, rate limiting.



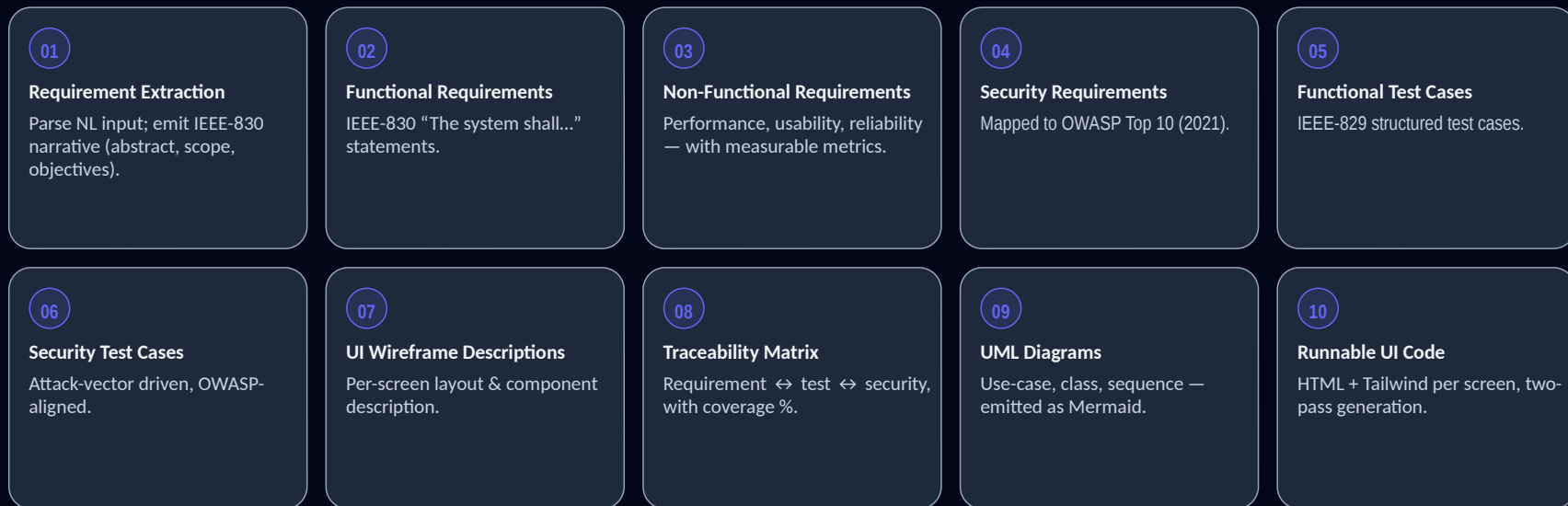
DATA

PostgreSQL on Neon

Serverless Postgres. Every artifact persisted keyed by (project, type) for resumability.

The 10-stage pipeline

From natural language to runnable artifacts.



Stages 3–5 parallel · 6–7 parallel · 8–10 parallel



Stages 1–9 · Groq llama-3.3-70b



Stage 10 · Google Gemini 2.5-flash-lite

Two LLMs, one pipeline

Each model owns what it's best at.

Groq

llama-3.3-70b-versatile

- Drives stages 1–9 — narrative, requirements, tests, traceability, UML as Mermaid.
- Fast inference on Groq's LPU tier keeps the pipeline responsive.
- Structured JSON output for downstream parsing & validation.
- Cost-efficient for high-volume text generation.

Google Gemini

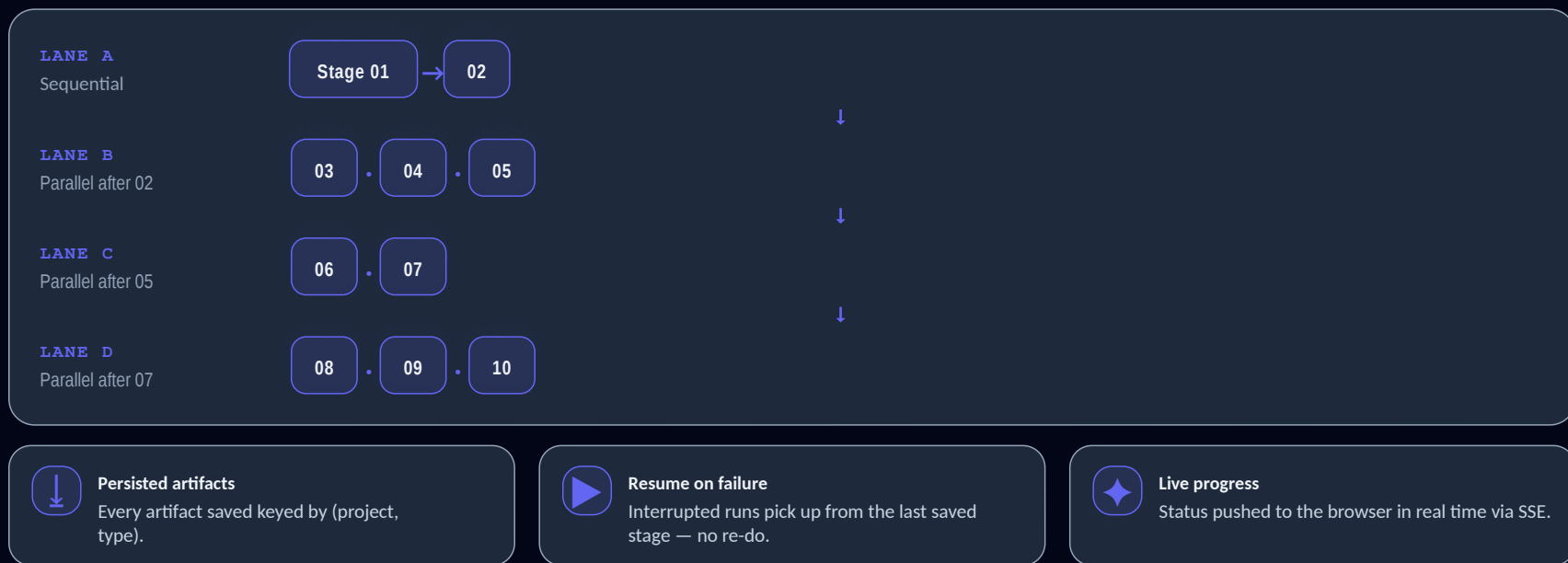
2.5-flash-lite

- Drives stage 10 — runnable UI code generation.
- Two-pass strategy: shared design system first, then one HTML page per distinct screen.
- Strong code-generation quality at a fraction of the cost of larger models.
- Produces Tailwind-styled, runnable HTML out of the box.

Pipeline router dispatches each stage to its assigned model — no manual handoff.

Parallel stages, resumable runs

Independent stages run concurrently; failures pick up where they stopped.



Security, not bolted on

OWASP Top 10 (2021) requirements auto-generated in stage 4; auth stack hardened end-to-end.



STAGE 4

OWASP-mapped requirements

Stage 4 auto-generates security requirements aligned to OWASP Top 10 (2021).



JWT access tokens

15-minute TTL; signed and verified on every request.



Rotating refresh tokens

7-day TTL, SHA-256 hashed, with token-family theft detection.



bcrypt password hashing

Cost factor 12; salts managed by bcrypt.



Account lockout

5 failed logins → temporary lockout + audit entry.



Helmet + CORS

Security headers + strict CORS allow-list.



Zod input validation

Every API route validates its payload with Zod schemas.



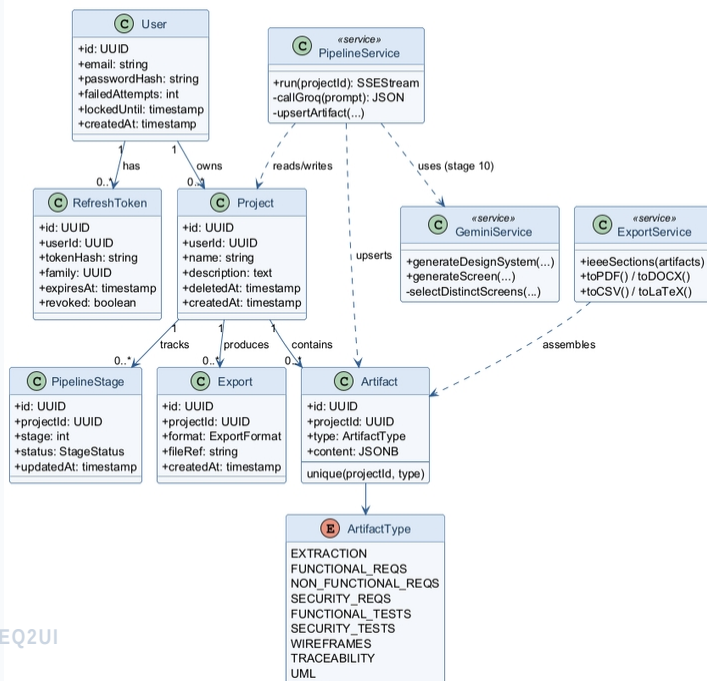
Rate limiting

Auth + generation endpoints throttled to prevent abuse.

Diagrams and runnable UI

Mermaid → PNG for UML; HTML + Tailwind per screen for UI.

Req2UI - Class Diagram (domain + services)



UML Use-case · class · sequence

- Use-case, class, and sequence diagrams emitted as Mermaid.
- Rasterized to PNG via [resvg-js](#) for embedding in PDF exports.

UI CODE (STAGE 10) Two-pass strategy

- Pass 1 — generate a shared design system: colors, typography, spacing, components.
- Pass 2 — emit one HTML + Tailwind page per distinct screen, all consuming the design system.
- Result — a low-fidelity but consistent and runnable UI prototype.

Every artifact links to every other

Illustrative matrix with coverage %, plus four export targets.

REQUIREMENT	TEST CASE	SECURITY	STATUS
FR-01 User login	TC-001	—	✓
SR-01 OWASP A07	—	STC-005	✓
FR-04 Export PDF	TC-018	—	✓
NFR-02 Response < 500ms	TC-022	—	✓
SR-03 OWASP A03	—	STC-009	✓



Example coverage: 100% of FRs mapped to a test case; 100% of SRs mapped to a security test.



PDF

Stakeholder-ready, paginated artifact bundle.



DOCX

Editable Word doc for downstream revision.



LaTeX

Academic / publication-ready source.



CSV

Machine-readable matrix for tooling.

Built on a modern TypeScript stack

One language end-to-end — frontend, backend, and tooling.



FRONTEND

React 19

TypeScript

Tailwind CSS v4

Vite

React Router 7



BACKEND

Node.js

Express 4

TypeScript

JWT

Zod

Helmet

bcrypt



DATA & AI

PostgreSQL (Neon serverless)

Groq llama-3.3-70b

Google Gemini 2.5-flash-lite



EXPORT & DEPLOY

pdfkit

docx.js

csv-writer

resvg-js

Vercel (FE)

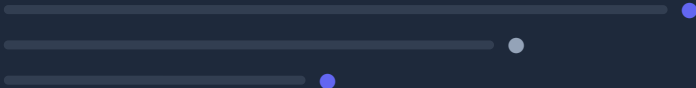
Render (BE)

Inside the app

Four screens the committee can ask to see live.

SCREEN 01

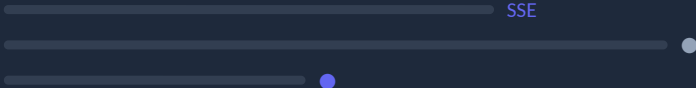
 **Dashboard**



- List of projects, status badges, last-run timestamp.
- Quick actions: New project, Open, Delete.

SCREEN 02

 **Project Detail**



SSE

- The 10 pipeline stages with live SSE progress.
- Re-run or resume from any saved stage.

SCREEN 03

 **Artifacts Viewer**



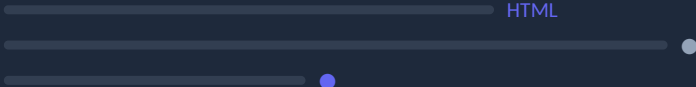
PDF

CSV

- Every generated artifact — SRS, tests, UML, UI code — listed.
- Each artifact individually downloadable.

SCREEN 04

 **UI Preview**



HTML

- Rendered HTML+Tailwind output per screen.
- Opened in a sandboxed iframe for live preview.

Validated, not just demoed

Automated tests + standards conformance across every artifact type.

24



AUTOMATED TESTS PASSING

Across 2 suites: auth routes + export service.

IEEE 830



SRS CONFORMANCE

Generated SRS follows IEEE-830 structure.

IEEE 829



TEST-DOC CONFORMANCE

Test cases use IEEE-829 fields.

OWASP Top 10 (2021)



SECURITY MAPPING

Every SR linked to a Top-10 category.

IEEE 12207



LIFECYCLE ALIGNMENT

Process standard followed end-to-end.

UML



DIAGRAM STANDARD

Use-case, class, sequence per UML spec.

What we delivered

Five objectives, five outcomes — all met.

01

Single input — Accepts free-form plain-language project descriptions.

✓ Met

02

Full SRS — IEEE-830 SRS with functional, non-functional, and security requirements.

✓ Met

03

Test cases — Functional tests (IEEE 829) + security tests (OWASP-driven).

✓ Met

04

UML + UI — Mermaid UML diagrams + HTML/Tailwind per screen.

✓ Met

05

Traceability + exports — Matrix with coverage % + PDF/DOCX/LaTeX/CSV.

✓ Met

Research gap closed: first integrated NL → SRS → UI → functional + security tests pipeline with auto security & full traceability.

Honest limits, clear roadmap

What we know is bounded — and where we're going next.



LIMITATIONS

- Output quality depends on the LLM and on the detail of the user's input.
- Throughput is bounded by API rate limits (free tiers on Groq & Gemini).
- Generated UI is a low-fidelity prototype — not production-ready.
- Some non-functional targets (load testing, full WCAG) still need dedicated test runs.
- Coverage of OWASP is Top-10-only; ASVS-level depth is future work.



FUTURE WORK

- Admin dashboard for project + user management.
- In-app artifact editing and refinement.
- Self-critique refinement loops (LLM reviews its own output).
- More export targets (e.g. Markdown, OpenAPI).
- Domain fine-tuning / retrieval augmentation.
- ASVS-level security requirements.
- Auto-run generated tests against generated UI.

Req2UI closes the gap.

The first integrated pipeline covering natural language → SRS → UI prototype → functional + security test cases — with automatic security generation and full end-to-end traceability.

ONE INPUT. TEN STAGES. A COMPLETE, SECURITY-AWARE ARTIFACT SUITE.

IEEE 830

IEEE 829

OWASP Top 10 (2021)

Thank you.

Questions are welcome.

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